

QUEUE

Q3) Delete an Element in a Queue (Queue Dequeue Simulation)

DS	Session	Queue	Question Information	Level 1	Challenge 53
Question description Ramesh is an DS expert training youngsters struggling in DS to make them better. Ramesh usually gives interesting problems to the youngsters to make them love the DS. One such day Sathya provided to the youngsters to solve the task such that, delete an element in a Queue, Queue data structures work on the FIFO architecture so the element that has entered first in the list will go out from the list first. Youngsters were lacking the idea to solve the problem. Being an exciting youngster can you solve it? Function Description					

1. Create a main function for the program.
2. In this main function accept the size and data of the queue by the user.
3. After that enqueue these element in the queue using an enqueue() function
4. Now to dequeue, we will be using an function that will remove elements from the queue one by one.
5. In this dequeue function first we will check if there are elements present in the queue or not.
6. If there are elements present in the queue then we will move the front of the queue by one step each time ,that is every time when an element is removed from the list.

CODE:

```

C main.c > enqueue
1  #include <stdio.h>
2
3  #define MAX 1000
4
5  int queue[MAX];
6  int front = 0, rear = -1;
7
8  void enqueue(int x) {
9      queue[++rear] = x;
10 }
11
12 void dequeue(void) {
13     if (front <= rear) {
14         front++;
15     }
16 }
17
18 void printQueue(void) {
19     for (int i = front; i <= rear; i++) {
20         printf(format: "%d%s", queue[i], (i == rear) ? " : " : " ");
21     }
22     printf(format: "\n");
23 }
24
25 int main(void) {
26     int n;
27     if (scanf(format: "%d", &n) != 1) return 0;
28
29     for (int i = 0; i < n; i++) {
30         int val;
31         scanf(format: "%d", &val);
32         enqueue(x: val);
33     }
34
35     printf(format: "Dequeuing elements:\n");
36     while (front < rear) {
37         dequeue();
38         printQueue();
39     }
40
41     return 0;
42 }

```

OUTPUT:

```

[suraj@suraj-83gs test]$ ./main
5
5 6 3 1 2
Dequeuing elements:
6 3 1 2
3 1 2
1 2
2
[suraj@suraj-83gs test]$

```